

DUBAI INTL, United Arab Emirates

WMO: 411940

Lat: 25.255N

Lon: 55.364E

Elev: 34

StdP: 14.68

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	56.8	58.7	31.9	26.3	86.0	35.8	30.8	84.0	22.3	69.8	20.3	69.3	4.9	180	0.443

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.6	109.9	74.4	107.7	74.9	105.7	75.5	86.7	95.7	85.7	95.0	84.8	94.7	10.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
84.5	181.1	92.9	83.6	176.0	92.7	82.4	169.0	92.4	51.7	96.3	50.5	94.8	49.4	94.9	90.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.8	17.7	16.1	53.5	115.3	2.5	1.5	51.7	116.4	50.3	117.3	48.9	118.2	47.1	119.3		
(5)			46.6	88.7	2.7	0.9	44.7	89.4	43.1	90.0	41.6	90.5	39.6	91.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	84.0	68.1	70.5	75.2	82.7	90.2	94.1	97.6	98.0	93.5	87.3	78.8	71.5		
	DBStd	11.31	3.62	5.03	5.68	5.86	4.65	3.91	3.92	3.13	3.10	3.19	3.80	3.72		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	17	10	5	0	0	0	0	0	0	0	0	0	2		
	CDD50	12425	560	575	783	981	1245	1322	1476	1489	1305	1157	865	667		
	CDD65	6966	105	159	318	531	780	872	1011	1024	855	692	415	204		
	CDH74	98124	345	804	2465	6371	11740	14021	16928	17108	13487	9717	4097	1041		
	CDH80	62860	33	205	872	3203	7499	9716	12465	12644	9175	5501	1415	132		
(14) Wind	WSAvg	8.1	7.6	8.4	8.7	8.3	8.6	8.5	8.8	8.6	7.9	7.4	7.4	7.2		
(15) Precipitation	PrecAvg	3.9	0.9	0.7	0.9	0.2	0.0	0.0	0.1	0.0	0.0	0.1	0.3	0.7		
	PrecMax	11.7	4.4	3.9	6.1	1.3	0.0	0.1	1.3	0.1	0.2	2.0	1.6	5.6		
	PrecMin	0.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	2.9	1.1	0.9	1.4	0.3	0.0	0.0	0.3	0.0	0.0	0.4	0.4	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.5	91.4	96.9	103.6	108.7	111.3	113.2	113.1	108.0	102.0	93.6	86.2		
		MCWB	63.8	64.2	64.9	67.8	69.8	72.5	74.4	73.8	72.9	70.5	66.8	65.6		
	2%	DB	80.5	85.7	91.8	99.0	105.5	107.7	110.2	109.7	105.4	98.7	91.1	82.8		
		MCWB	63.0	63.2	64.8	67.3	70.6	73.6	75.5	75.1	74.0	71.6	69.3	66.7		
	5%	DB	77.4	82.1	88.0	96.5	102.4	104.9	107.5	107.4	102.4	96.7	88.5	80.9		
		MCWB	63.6	63.6	64.8	67.0	70.5	74.7	76.5	75.9	75.1	72.8	70.2	66.9		
	10%	DB	75.3	78.8	84.5	93.1	100.1	102.3	104.8	104.6	100.5	94.8	86.4	79.1		
		MCWB	63.4	63.6	64.7	67.5	71.5	75.7	77.4	77.2	76.4	74.1	70.6	66.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.2	70.4	72.7	76.5	82.4	86.2	88.0	88.2	86.4	84.0	77.6	72.8		
		MCDB	76.0	77.6	81.2	87.4	92.6	95.5	97.1	96.6	94.0	91.8	86.0	79.5		
	2%	WB	67.6	68.8	71.1	74.7	80.3	84.8	86.7	86.8	85.4	82.4	75.5	71.2		
		MCDB	75.0	76.4	79.5	86.3	91.2	94.3	96.2	95.8	93.4	90.1	84.8	78.7		
	5%	WB	66.3	67.5	69.9	73.1	78.8	83.8	85.6	85.6	84.4	81.0	74.3	69.7		
		MCDB	74.1	75.5	78.7	85.3	90.7	93.6	95.6	95.2	93.0	89.4	83.8	77.5		
	10%	WB	65.1	66.4	68.8	71.9	77.3	82.6	84.6	84.6	83.5	79.8	72.9	68.5		
		MCDB	73.0	74.9	77.8	83.9	90.5	93.3	95.4	95.2	92.5	88.9	82.8	76.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.5	15.3	16.1	18.3	19.6	18.5	16.3	16.6	16.7	16.9	15.5	14.8		
		MCDBR	17.3	19.0	20.7	21.9	22.2	21.0	18.7	18.9	18.6	18.6	17.2	16.7		
		MCWBR	8.2	8.8	10.1	10.5	10.9	11.9	11.6	11.5	11.8	11.1	8.7	8.3		
	5% WB	MCDBR	15.1	15.6	16.4	18.7	20.2	18.5	15.8	16.4	16.5	16.6	16.1	15.3		
		MCWBR	8.5	8.7	8.8	10.1	11.0	10.7	10.1	10.5	10.3	9.6	8.4	8.4		
(40) Clear-Sky Solar Irradiance	taub		0.443	0.502	0.563	0.595	0.588	0.637	0.726	0.689	0.599	0.514	0.457	0.438		
	taud		2.088	1.918	1.774	1.731	1.742	1.687	1.569	1.638	1.773	1.934	2.084	2.105		
	Ebn at Noon		252	245	238	234	235	222	203	210	226	238	246	247		
	Edh at Noon		46	58	70	75	74	78	87	81	69	56	45	43		
(44) All-Sky Solar Radiation	RadAvg		1283	1534	1814	2094	2350	2395	2255	2137	1980	1723	1390	1214		
	RadStd		83	122	124	128	51	46	60	48	29	25	42	81		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.62	+2.36	N/A	+0.99	N/A	N/A	-9	+623	+617
(47) Regional (15 neighbors)		+0.85	+1.26	N/A	N/A	N/A	N/A	N/A	+270	+268

Nomenclature: See separate page